

United International University

Department of Computer Science and Engineering

Complete Legal Aid

An AI-Powered Legal Tech Platform for Accessible Justice in Bangladesh

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Section: D
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Project Repository

`https://github.com/Tanjamul-Azad/Complete-Legal-Aid.git`

Table of Contents

Contents

Program Outcomes **3**

Abstract **4**

1 Introduction **5**

2 Motivation **5**

3 Objectives of the System **5**

4 System Overview **6**

5 Feature Description **6**

5.1 Welcome Page 6

5.1.1 Core Features 7

5.2 AI Legal Assistant 7

5.3 Emergency Legal Help 8

5.4 Authentication System 8

5.5 Citizen Dashboard 9

5.6 Legal News Module 9

5.7 Lawyer Discovery 10

5.8 Case Management System 11

5.9 Blogs and Legal News 12

5.10 Online Payment System 12

5.11 Lawyer Dashboard 13

5.12 Admin Dashboard 13

6 Technology Stack **14**

6.1 Architecture Overview 14

6.2 Frontend Technologies 14

6.3 Backend Technologies 15

6.3.1 Backend File Structure 16

6.4 Database Architecture 16

6.5 AI and Machine Learning Integration 16

6.6 Real-Time Communication Layer 17

6.7 Authentication and Security Architecture 17

6.8 External Integrations 17

6.9 DevOps and Deployment Architecture 17
6.10 Development and Testing Tools 18
6.11 Key Technical Highlights 18

7 System Testing and Validation **18**

8 Challenges **19**

9 Future Enhancements **19**

10 Conclusion **20**

Mapped Program Outcomes

Table 1: Program Outcome (PO) Mapping for the Project

Program Outcome (PO)	**Project Alignment**
PO2: Problem Analysis	The project identifies critical legal accessibility challenges in Bangladesh, including high service costs, procedural complexity, and limited public awareness of legal rights. By analyzing these barriers, a technical solution was formulated in the form of an AI-assisted legal aid platform that provides structured guidance and simplified access to legal resources. This demonstrates the ability to analyze real-world problems and translate them into computable system requirements.
PO3: Design and Development of Solutions	The system was architected to address specific societal needs through a secure, scalable legal aid platform. Complex features such as a dual-AI architecture (Gemma and Gemini), a secure document vault, and role-based dashboards for citizens, lawyers, and administrators were designed and implemented with strong emphasis on privacy, security, and usability.
PO5: Modern Tool Usage	Modern engineering tools and frameworks were extensively used throughout the project. The frontend was developed using React 19.2 with TypeScript, while the backend was implemented using Django and Django REST Framework. Advanced AI technologies were integrated for legal assistance. Git and GitHub were used for version control, collaboration, and project management.
PO8: Ethics	Professional and ethical responsibility were core considerations, particularly in handling sensitive legal information. JWT-based authentication, strict role-based access control, and encrypted communication channels were implemented to ensure confidentiality and secure citizen–lawyer interactions.
PO9: Individual and Teamwork	The project was completed by a four-member team operating under the name "Deadliners." Responsibilities were distributed across frontend development, backend services, database management, and AI integration, demonstrating effective collaboration, coordination, and accountability.
PO10: Communication	Effective communication was demonstrated through comprehensive documentation and technical reporting. System architectures, workflows, and implementation details were presented clearly for both technical and non-technical audiences.
PO12: Lifelong Learning	The development process required independent exploration of emerging technologies, including fine-tuning large language models using LoRA and implementing real-time data synchronization with WebSockets. This reflects the ability to adapt to new tools and pursue continuous learning beyond formal coursework.

Abstract

Access to legal services in Bangladesh remains limited due to a lack of awareness, complex legal procedures, financial constraints, and the absence of reliable digital platforms. This report presents *Complete Legal Aid (CLA)*, a full-stack web-based legal assistance platform designed to connect citizens with verified lawyers, provide AI-powered legal guidance, enable secure case and document management, and support emergency legal reporting.

The system integrates modern frontend and backend technologies with a dual artificial intelligence architecture using the Gemini API and a fine-tuned Gemma language model trained on Bangladesh-specific legal data. CLA demonstrates the practical application of software engineering principles to build a scalable, secure, and citizen-centric legal technology solution.

1. Introduction

Legal assistance is a fundamental requirement to ensure justice and protect citizens' rights. However, in Bangladesh, access to legal services is often limited by high consultation costs, lack of legal awareness, geographical limitations, and difficulty in identifying trustworthy legal professionals. Traditional legal workflows are highly reliant on physical interactions and paper-based documentation, which limits efficiency and transparency.

With the growth of digital infrastructure, there is an opportunity to modernize the legal service ecosystem. *Complete Legal Aid (CLA)* was developed to address these challenges by providing a centralized digital platform where citizens can access legal information, consult verified lawyers, manage legal cases, and securely store sensitive documents.

The project follows core software engineering principles such as modularity, scalability, maintainability, and security, making it suitable for real-world deployment and future enhancement.

2. Motivation

Legal services in Bangladesh remain largely inaccessible to a significant portion of the population due to high costs, lack of legal awareness, complex procedures, and dependence on traditional paper-based systems. Many citizens are unsure about their legal rights or how to initiate legal action, especially during emergency situations. In addition, identifying trustworthy and verified legal professionals is often time-consuming and challenging.

The motivation behind developing **Complete Legal Aid (CLA)** arises from the need to modernize the legal service ecosystem through a digital platform that is accessible, transparent, and efficient. With the increasing availability of internet access and smartphones, there is an opportunity to introduce technology-driven solutions that simplify legal processes and reduce barriers to justice.

Another key motivation is the absence of jurisdiction-aware AI-based legal assistance platforms tailored specifically for Bangladesh. Most existing legal AI tools lack contextual understanding of local laws. By integrating a fine-tuned language model trained on Bangladesh-specific legal data, the CLA aims to provide more relevant and responsible legal guidance.

In general, the project is motivated by the goal of empowering citizens, improving legal awareness, and contributing to the digital transformation of legal services in Bangladesh.

3. Objectives of the System

The primary objective of the **Complete Legal Aid (CLA)** project is to design and develop a secure, scalable, and user-centric digital legal assistance platform that improves access to legal services in Bangladesh. The system aims to bridge the gap between citizens and verified legal professionals by leveraging modern web technologies and artificial intelligence.

The specific objectives of the project are as follows:

- Develop a complete web-based legal assistance system using modern software engineering practices.

- To provide citizens with easy access to verified lawyers based on specialization and case requirements.
- To offer AI-powered preliminary legal guidance to improve legal awareness among the general public.
- To enable secure digital case creation, document upload, and case progress tracking.
- To support anonymous emergency legal reporting for sensitive and time-critical situations.
- To ensure role-based access control for citizens, lawyers, and administrators.
- Demonstrate the practical application of software engineering principles in solving real-world legal accessibility problems in Bangladesh.

4. System Overview

CLA is a role-based web application designed for three primary user roles:

- **Citizen**: Individuals seeking legal assistance.
- **Lawyer**: Verified legal professionals providing consultation and case handling.
- **Administrator**: System managers responsible for verification and monitoring.

The system follows a client–server architecture where a React-based frontend communicates with a Django REST backend via secure APIs. Persistent data, such as users, cases, documents, payments, and logs are stored in a relational MySQL database. Artificial intelligence services are integrated to improve legal guidance and user support.

5. Feature Description

This section describes the core functional features of the Complete Legal Aid platform along with their user interface representations.

5.1 Welcome Page

The welcome page serves as the public entry point of the system. It introduces the objectives of the platform, highlights key features, explains how the system works, and displays daily legal news collected from trusted sources.

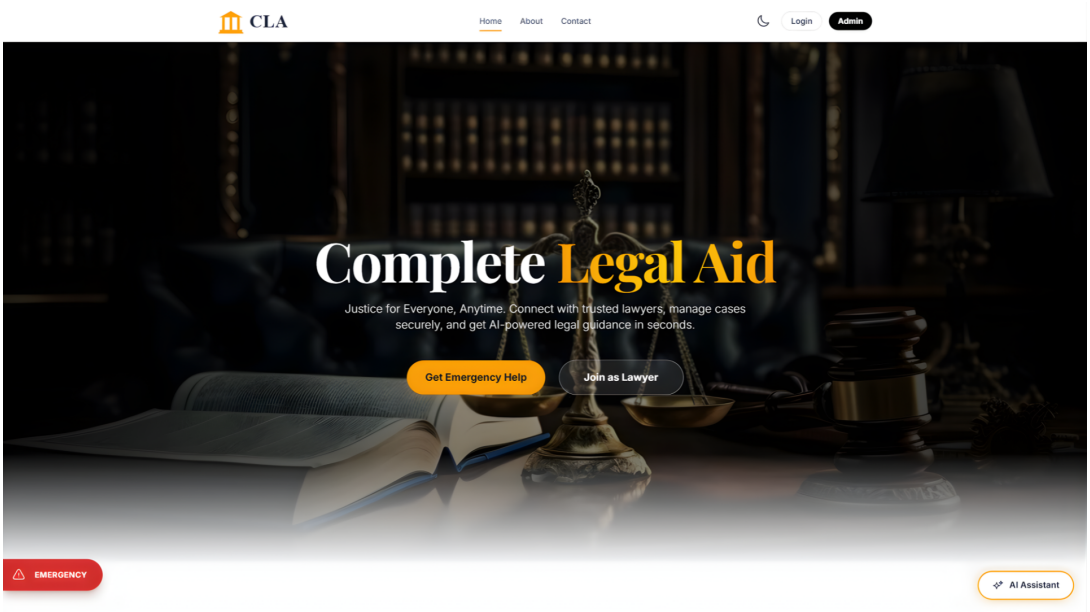


Figure 1: Landing Page Overview

5.1.1 Core Features

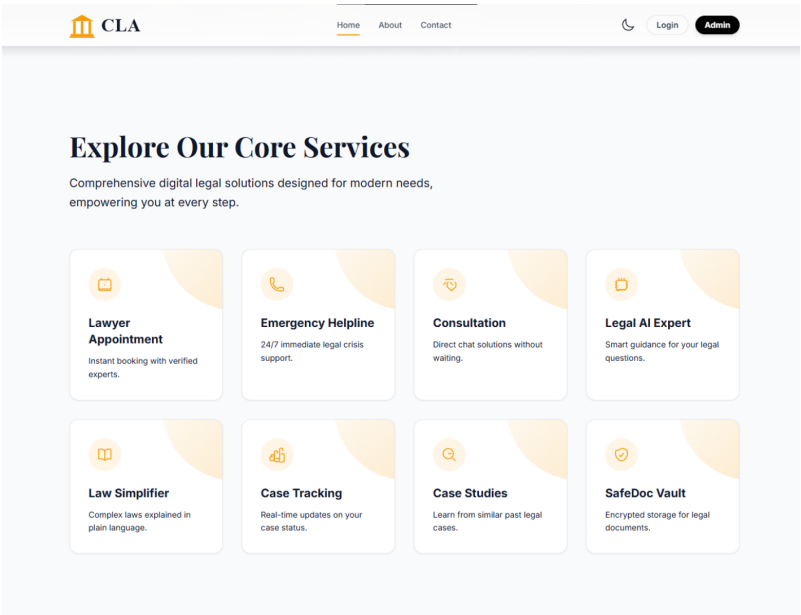


Figure 2: Features

5.2 AI Legal Assistant

The AI Legal Assistant provides preliminary legal guidance in Bangla and English. It is accessible without login and helps users understand legal concepts and procedures before consulting a lawyer.

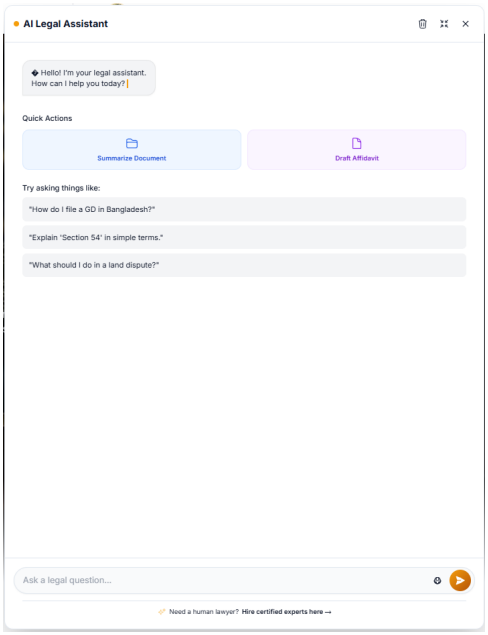


Figure 3: AI Legal Assistant Interface

5.3 Emergency Legal Help

This feature allows users to submit urgent legal reports anonymously. It is designed to encourage reporting of sensitive and time-critical legal issues while ensuring user safety.

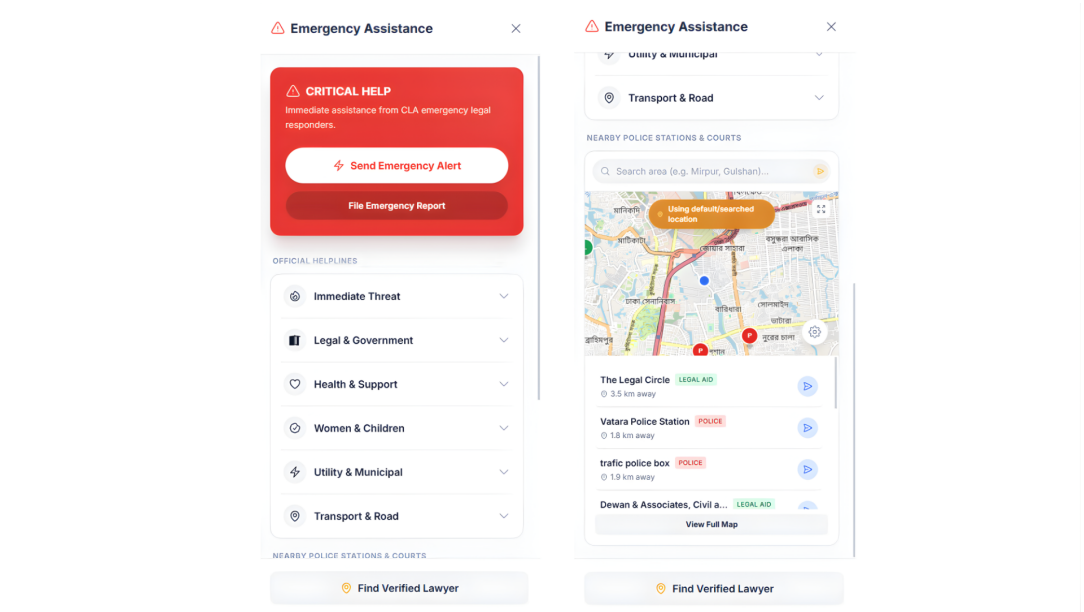


Figure 4: Emergency Legal Help Interface

5.4 Authentication System

CLA uses JWT-based authentication to ensure secure login and role-based access control. Each user role has controlled access to system resources.

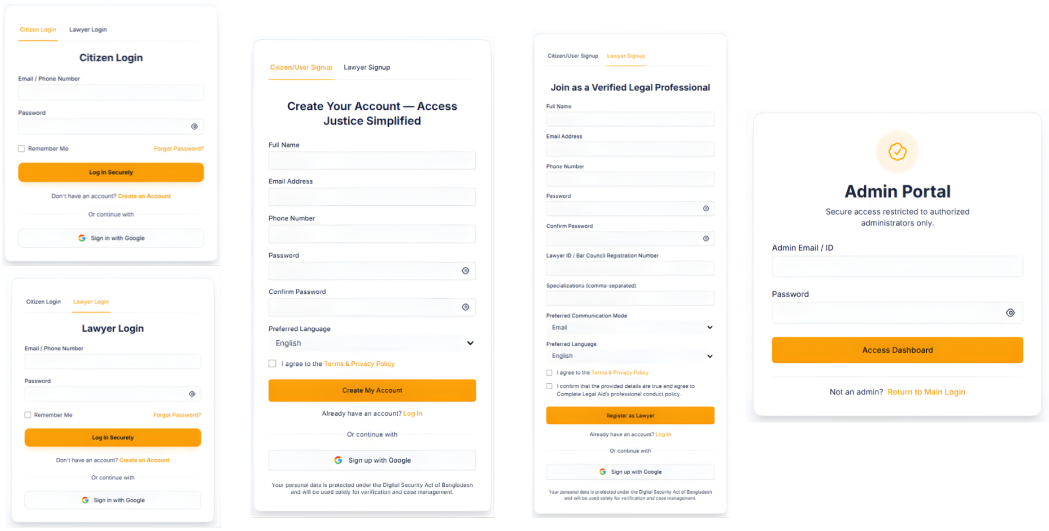


Figure 5: User Authentication Interface

5.5 Citizen Dashboard

After authentication, citizens are redirected to a personalized dashboard that displays active cases, consultation bookings, notifications, and shortcuts to key features.

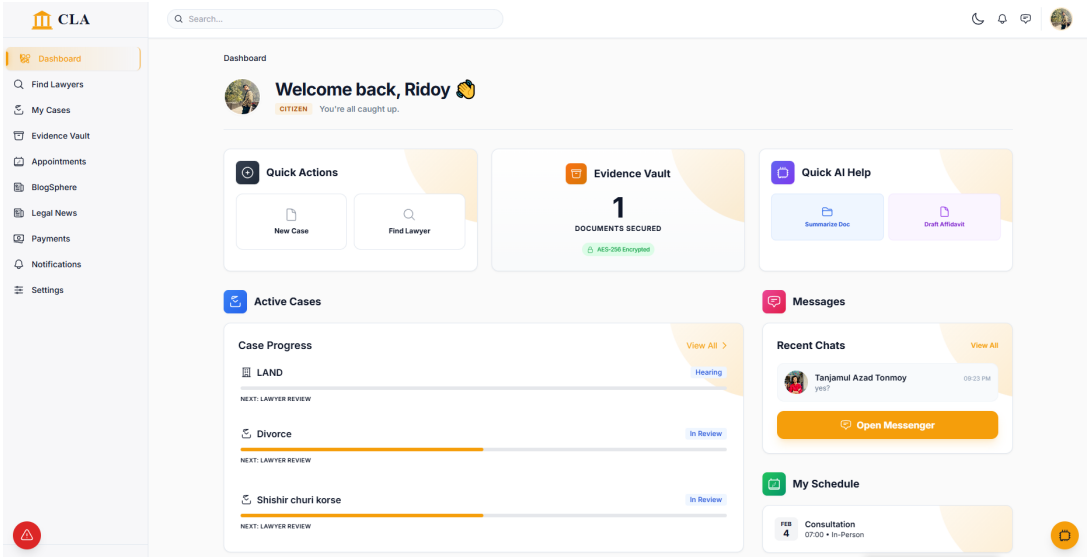


Figure 6: Citizen Dashboard

5.6 Legal News Module

- **AI-Curated News:** Displays up-to-date legal news related to law, justice, and legal developments.
- **Relevance Scoring:** Each article is ranked using AI-based relevance analysis.
- **Confidence Indicator:** Shows an AI-generated confidence score for improved reliability.

- **Trusted Sources:** News is collected from reliable sources such as BBC, CNN, Al Jazeera, and NPR.
- **Quick Access:** Users can read summaries and access full articles via external links.

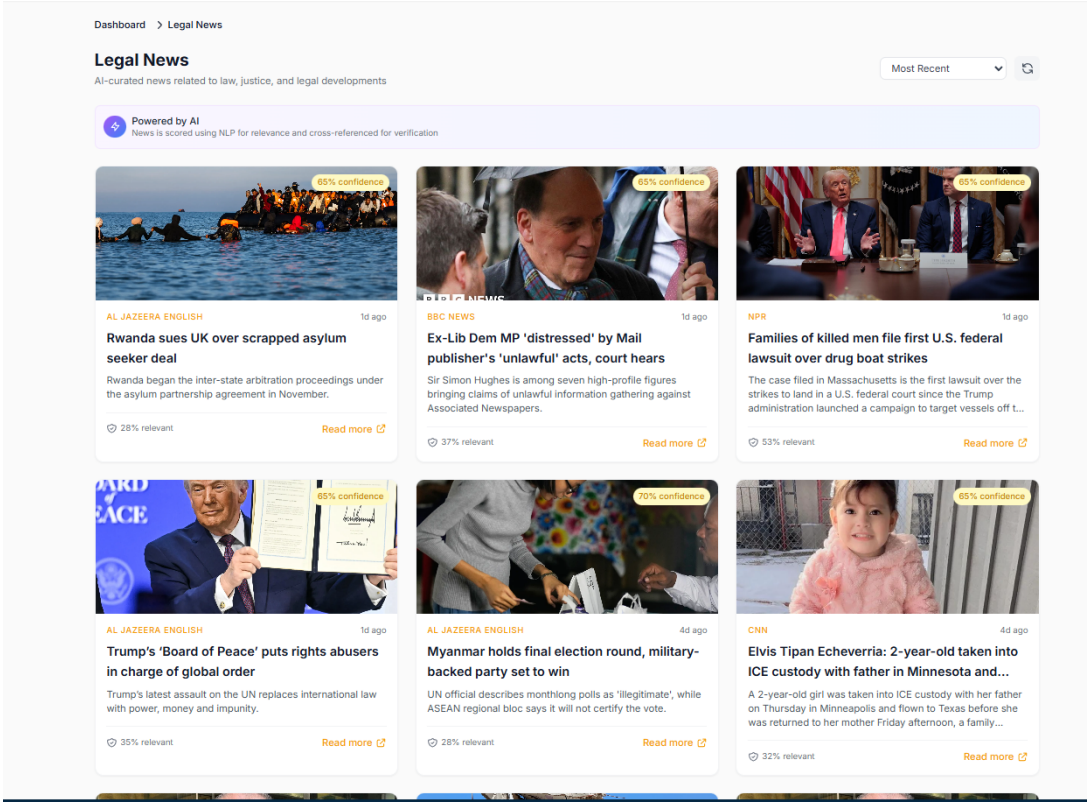


Figure 7: Legal News Insight

5.7 Lawyer Discovery

Citizens can browse and search verified lawyers based on specialization and profile information. This feature promotes transparency and informed decision-making.

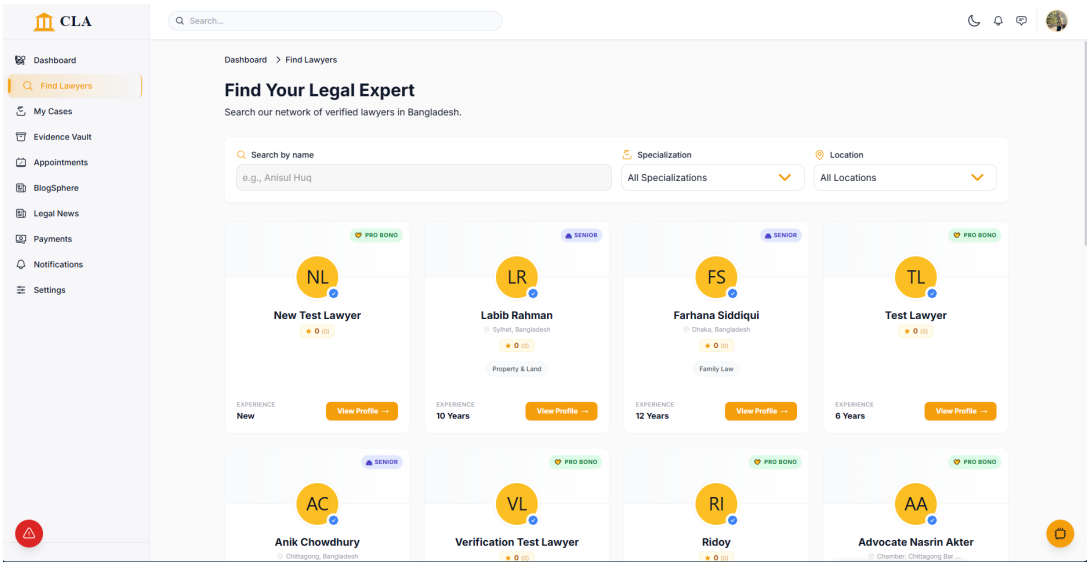


Figure 8: Lawyer Discovery Interface

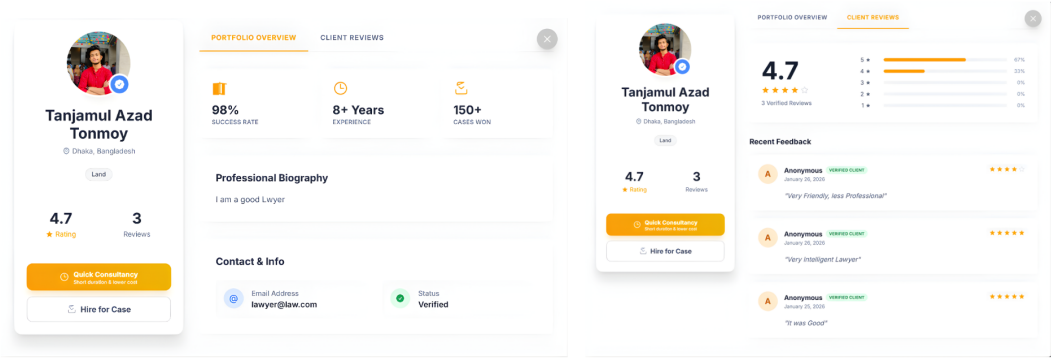


Figure 9: Lawyer Profile View

5.8 Case Management System

The case management system allows citizens to create legal cases, upload documents, and track progress through defined stages. Lawyers can review and update assigned cases.

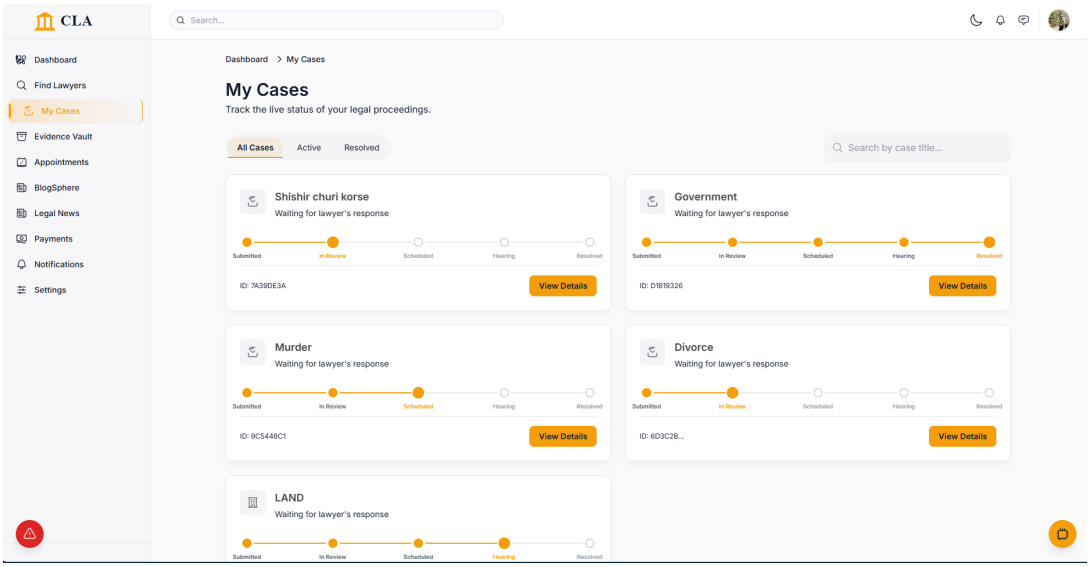


Figure 10: Case List View

5.9 Blogs and Legal News

Lawyers can publish educational blog posts, while the legal news section provides up-to-date legal information aggregated from reliable sources.

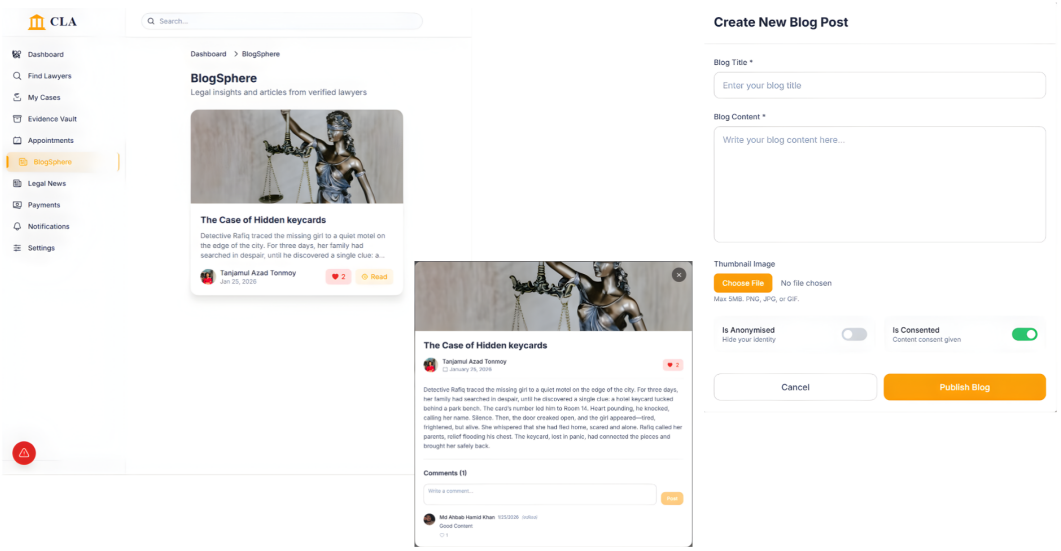


Figure 11: Legal Blogs Interface

5.10 Online Payment System

The platform includes an implemented payment module that enables secure payment for consultations and case-related services.

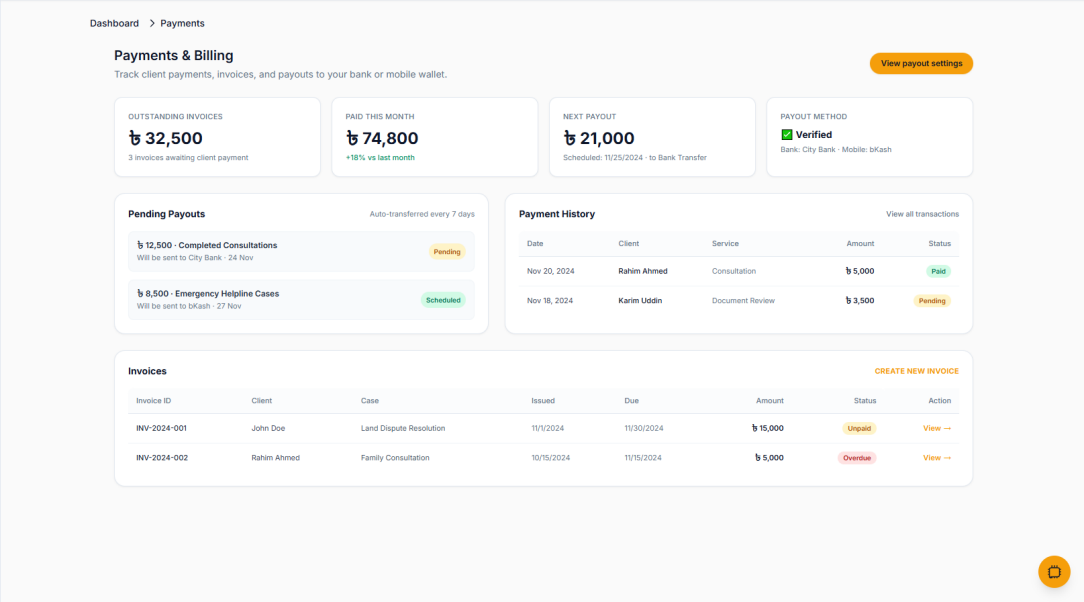


Figure 12: Online Payment Interface

5.11 Lawyer Dashboard

The lawyer dashboard enables legal professionals to manage cases, handle booking requests, and communicate with clients efficiently.

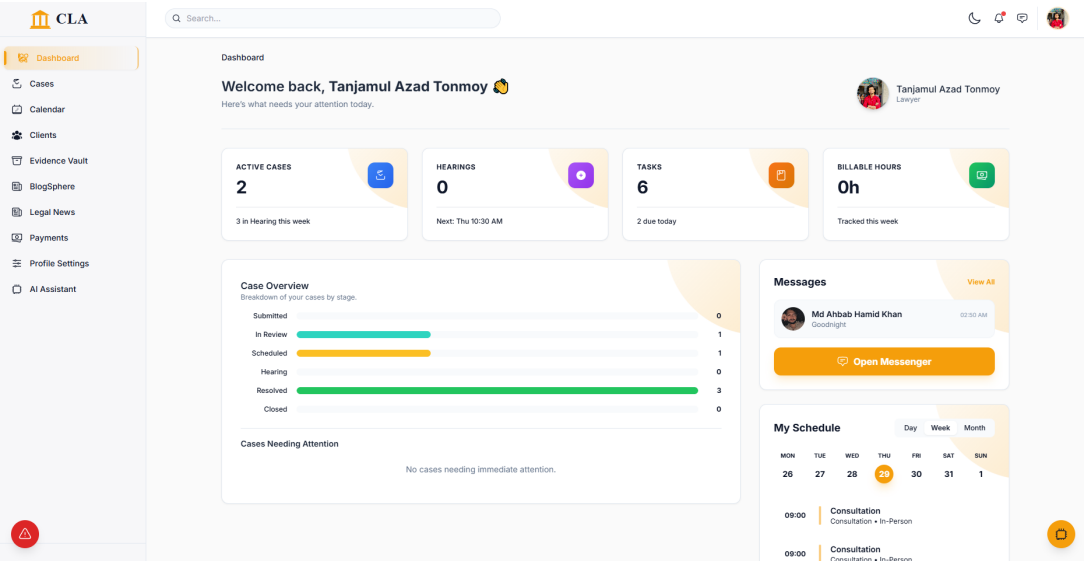


Figure 13: Lawyer Dashboard

5.12 Admin Dashboard

Administrators manage user accounts, verify lawyers, and monitor system activities through the admin dashboard.

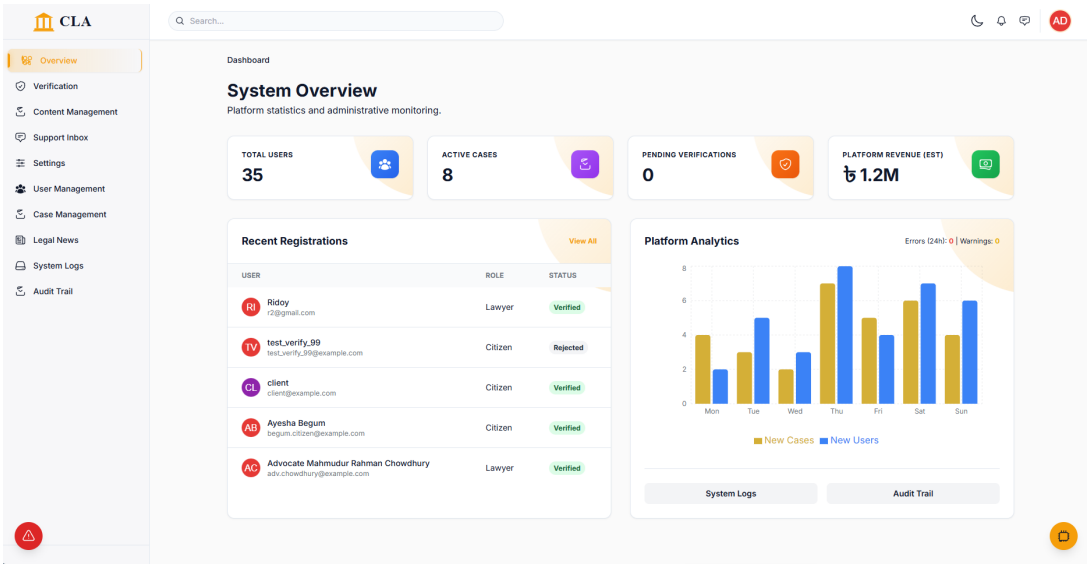


Figure 14: Admin Dashboard

6. Technology Stack

This section presents a comprehensive overview of the technology stack used in the development of the **Complete Legal Aid (CLA)** platform. The system is designed using a modular, scalable, and production-ready architecture that integrates modern frontend technologies, a robust backend framework, a relational database, artificial intelligence models, real-time communication, and secure deployment practices.

6.1 Architecture Overview

CLA follows a layered client-server architecture. The frontend communicates with the backend through RESTful and WebSocket-based APIs. The backend handles business logic, authentication, AI orchestration, and database operations, while external services support AI inference, authentication, and data aggregation.

This architecture ensures separation of concerns, scalability, and ease of maintenance.

6.2 Frontend Technologies

The frontend is implemented as a single-page application (SPA) using modern JavaScript tooling to ensure performance, maintainability, and developer productivity.

- **React 19.2**: Used for building reusable, component-based user interfaces.
- **TypeScript 5.8**: Ensures static type safety, improved code quality, and better tooling support.
- **Vite 6.2**: Provides fast development server startup and optimized production builds.
- **Tailwind CSS**: Enables utility-first styling with responsive and consistent UI design.

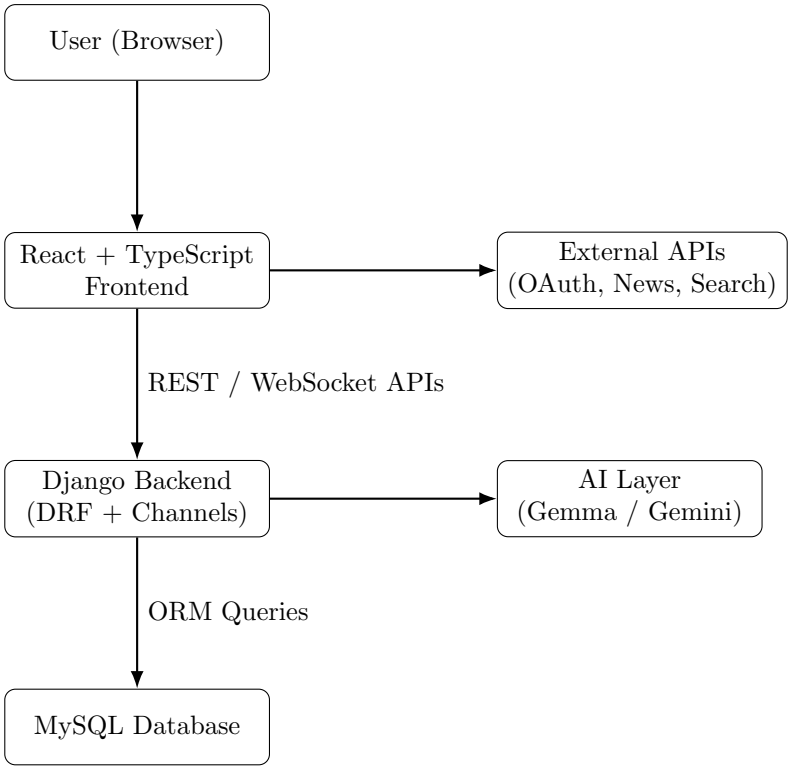


Figure 15: System Architecture Overview

- **Recharts**: Used for rendering analytical charts and dashboards for lawyers and administrators.
- **Axios**: Handles HTTP communication with backend REST APIs.
- **React Router**: Manages role-based client-side routing and navigation.

The frontend follows a feature-based folder structure to improve scalability and long-term maintainability.

6.3 Backend Technologies

The backend is developed using Django and Django REST Framework, providing a secure and scalable foundation for handling business logic and API services.

- **Django 4.2**: Core backend framework responsible for request handling, ORM, and security.
- **Django REST Framework (DRF)**: Used to expose RESTful APIs for frontend communication.
- **Daphne**: ASGI server used for handling asynchronous requests and WebSocket connections.
- **Django Channels**: Enables real-time communication via WebSockets.

6.3.1 Backend File Structure

```
backend/
|-- accounts/       % User management and roles
|-- cases/          % Legal case handling and workflow
|-- payments/       % Transactions and billing
|-- ai_services/    % AI request routing and inference
|-- notifications/  % Alerts and real-time updates
|-- core/           % Shared utilities and configurations
|-- settings/       % Environment-specific configurations
‘-- manage.py
```

This modular structure improves the separation of responsibilities and simplifies feature expansion.

6.4 Database Architecture

CLA uses MySQL 8.0 as the primary relational database to ensure structured storage, transactional integrity, and efficient querying. All database operations are abstracted through the Django ORM, ensuring schema consistency and portability.

The system contains over **30 normalized data models** grouped into the following logical domains:

- User identities, authentication, and role management
- Legal cases, case stages, and document metadata
- Payments, billing records, and transaction history
- Notifications, logs, and system activity tracking
- AI interaction history and query records

An Entity–Relationship model was designed to minimize redundancy while preserving relational integrity and query performance.

6.5 AI and Machine Learning Integration

CLA integrates a dual-model AI architecture to balance legal specialization with general-purpose intelligence.

The primary model, **Gemma 3**, is fine-tuned using **Low-Rank Adaptation (LoRA)** on Bangladesh-specific legal data to provide jurisdiction-aware responses. Model training and inference are managed using PyTorch and Hugging Face Transformers.

For unauthenticated users and general queries, the system dynamically routes requests to the **Gemini API** as a fallback large-scale model.

This routing strategy ensures both scalability and domain accuracy while optimizing computational cost.

6.6 Real-Time Communication Layer

Real-time system events such as notifications and status updates are handled using WebSockets powered by Django Channels. Redis acts as the channel layer, enabling message brokering across asynchronous workers.

This architecture allows future extensibility for features such as live chat, consultation sessions, and streaming updates without architectural changes.

6.7 Authentication and Security Architecture

Security is enforced at multiple layers of the system.

- JWT-based stateless authentication for API access
- Role-Based Access Control (RBAC) supporting Citizen, Lawyer, Admin, and Moderator roles
- Google OAuth 2.0 integration for secure third-party login
- HTTPS enforcement for encrypted communication
- Permission guards applied at both API and frontend routing levels

6.8 External Integrations

Several third-party services are integrated to enhance platform functionality.

- **Google OAuth**: Social login and identity verification.
- **Gemini API**: Large-scale AI inference.
- **NewsAPI**: Aggregation of daily legal news.
- **DuckDuckGo Search API**: Legal information discovery and enrichment.

6.9 DevOps and Deployment Architecture

The system follows a production-grade deployment model.

- Nginx as reverse proxy and static asset server
- Gunicorn (WSGI) and Daphne (ASGI) for handling mixed workloads
- Redis for caching and channel layer support
- Cloud storage for secure legal document management
- Environment variable configuration for secrets and deployment flexibility

6.10 Development and Testing Tools

To ensure code quality and maintainability, the following tools were used:

- Postman for API validation
- PyTest for unit and integration testing
- ESLint and Prettier for frontend code consistency
- Git and GitHub for version control and collaboration
- Markdown-based technical documentation (`TECH_STACK.md`)

6.11 Key Technical Highlights

- Modular full-stack architecture with clear separation of concerns.
- 30+ normalized database models with relational integrity.
- Dual AI pipeline combining fine-tuned and large-scale models.
- Real-time communication support using WebSockets.
- Production-ready deployment design with scalability considerations.

7. System Testing and Validation

System testing and validation were conducted to ensure the reliability, security, and correct functioning of the Complete Legal Aid platform. Both manual and API-based testing approaches were used to verify critical system workflows across different user roles.

The following testing activities were performed:

- **Authentication Testing:** Secure user registration, login, and logout functionalities were tested using JWT-based authentication. Role-based access control was verified to ensure authorized access.
- **Role-Based Feature Testing:** Citizen, lawyer, and admin dashboards were tested independently to confirm correct access permissions and feature availability.
- **Case Management Testing:** Case creation, document upload, case assignment, and case status update workflows were validated between citizens and lawyers.
- **API Testing:** RESTful APIs were tested using tools such as Postman to verify request–response behavior, data validation, and error handling.
- **AI Response Validation:** AI-generated legal guidance was manually reviewed to ensure relevance, clarity, and alignment with the Bangladesh-specific legal context.
- **Security Validation:** Protected endpoints were tested to confirm that unauthorized access was restricted and sensitive data remained secure.

These testing activities helped identify integration issues, improve system stability, and ensure consistent behavior across different system components.

8. Challenges

During the development of Complete Legal Aid (CLA), several technical and practical challenges were encountered:

- **AI Accuracy and Reliability:** Ensuring accurate and context-aware legal responses while minimizing hallucinations was a significant challenge.
- **Balancing AI Speed and Legal Precision:** Managing the trade-off between fast responses from the Gemini API and jurisdiction-specific accuracy from the fine-tuned Gemma model required careful design decisions.
- **Lawyer Verification Process:** Designing a reliable and secure lawyer verification mechanism while maintaining system usability required administrative oversight.
- **Frontend–Backend Integration:** Integrating the React frontend with the Django REST backend involved careful handling of authentication tokens and API synchronization.
- **Data Privacy and Security:** Managing sensitive legal documents and personal information requires strict access control and secure data handling practices.
- **Limited Legal Training Data:** Fine-tuning the Gemma model using Bangladesh-specific legal data was constrained by limited publicly available datasets.

9. Future Enhancements

Although Complete Legal Aid (CLA) delivers its core functionalities successfully, several enhancements can be implemented to further improve the platform:

- Integration of real-time video and audio consultation between citizens and lawyers.
- AI-based legal document summarization and automated document analysis.
- Development of a mobile application to increase accessibility for users in remote areas.
- Integration with NGOs and legal aid organizations to expand support coverage.
- Multilingual expansion to support additional local languages.
- Advanced analytics dashboards for lawyers and administrators.
- Potential integration with government services such as general diary tracking or court scheduling systems.

10. Conclusion

Complete Legal Aid (CLA) demonstrates the effective application of software engineering principles and artificial intelligence to improve legal accessibility in Bangladesh. By integrating secure workflows, AI-powered assistance, and role-based system design, the platform provides a scalable and practical solution for modern legal service delivery.